Teleoperating Robots Using Body-Machine Interface: Advancing Assistive Technology

Mohammed Najie AL Dreay
Dept. DIBRIS
University of Genoa
Genoa, Italy
Mohammednajie.aldreay@edu.unige.it

Camilla Pierella
Dept. DIBRIS
University of Genoa
Genoa, Italy
Camilla.pierella@unige.it

Matteo Moro
Dept. DIBRIS
University of Genoa
Genoa, Italy
Matteo.moro@unige.it

Fabio Rizzoglio
Dept. of Neuroscience
Northwestern University
Chicago, IL, United States of America
Fabio.rizzoglio@northwestern.edu

Maura Casadio
Dept. DIBRIS
University of Genoa
Genoa, Italy
Maura.Casadio@unige.it

***Abstract*—Body-machine interfaces (BoMIs) enable continuous teleoperation of assistive devices by translating body signals into control commands. Here, we focus on BoMIs based on body movements, where high-dimensional input data are mapped onto a low-dimensional control subspace. We investigate whether nonlinear mappings provide practical advantages over conventional linear approaches in navigation-like tasks. Specifically, we compare a linear mapping based on Principal Component Analysis (PCA) and a nonlinear mapping based on an Autoencoder (AE) for teleoperation of a differential-drive mobile robot base. Input signals consisted of a 42-dimensional vector of hand landmark coordinates extracted in real time from video captured by a standard camera and mapped to a 2D control command. Fourteen unimpaired adults participated in the experiment, divided into two groups: 7 using PCA and 7 using AE. Performance improved across test blocks in both groups, indicating learning under each mapping. However, the AE group consistently achieved faster completion times, fewer transitions between command regions of the control interface, and fewer collisions than the PCA group, while robot path length decreased with practice in both groups. These results indicate that, under strong dimensionality reduction, nonlinear AE mappings enable more efficient, stable, and safer robot teleoperation than linear PCA mappings, providing practical guidance for webcam-based assistive interfaces.**

***Index Terms*—Body–machine interface, teleoperation, mobile robot navigation, human–robot interaction, assistive robotics.**

I. Introduction

Body–machine interfaces (BoMIs) provide a link between human body signals and external device commands. In these interfaces, body movements are transformed into control variables such as cursor position or end-effector motion [1], [2]. In BoMIs that implement continuous control, body movements are transformed into command signals, allowing users to make smooth, ongoing corrections during task execution [3]. This makes them particularly relevant for teleoperation tasks, where the user must control a remote system in a stable, intuitive, and continuous manner [4], [5]. A central challenge in the design of BoMIs that implement continuous control is the definition of the user-to-device mapping. In many BoMIs, the user's body signals have higher dimensionality than the device command space; therefore, the interface must transform high-dimensional movements into a lower-dimensional control representation. Previous studies have explored both direct mappings and dimensionality-reduction approaches for controlling cursors, robotic arms, wheelchairs, telepresence robots, and drones [6]–[10]. Direct mappings can be simple and intuitive because specific body signals are assigned directly to device commands; for example, in an sEMG-based cursor-control interface, activation of the right and left biceps was mapped to right and left cursor motion, while activation of the trapezius muscles was mapped to vertical cursor motion [11]. However, these mappings often depend on predefined correspondences between body motions and device commands. In contrast, mappings based on dimensionality-reduction algorithms can extract control variables from spontaneous body movements and may better adapt to the structure of the user's motor signals [12]–[14]. Among dimensionality-reduction methods, Principal Component Analysis (PCA) has been widely used to identify low-dimensional control spaces that capture the main variance of natural body movements [6], [8], [15]. PCA provides a simple and interpretable linear mapping and has shown effective performance in cursor control and wheelchair-like navigation tasks. However, it cannot capture nonlinear relationships in body-motion data. Moreover, PCA distributes variance in a rank-ordered manner across components, which may produce an imbalanced contribution of the control dimensions when strong dimensionality reduction is required. More recently, nonlinear approaches based on Autoencoders (AEs) have been proposed to learn compact latent representations of body signals for continuous robot control [7]. These approaches may provide a more flexible mapping by preserving nonlinear structure in the input space while distributing information more evenly across the latent dimensions. Despite these advances, the comparative advantage of nonlinear AE-based mappings over conventional PCA-based mappings remains

This work was supported by the Italian Ministry of Research, under the complementary actions to the National Recovery and Resilience Plan (NRRP) "Fit4MedRob-Fit for Medical Robotics" Grant (# PNC0000007).

unclear in mobile robot teleoperation tasks. Existing studies differ in sensing modality, task design, controlled device, and performance metrics, making it difficult to isolate the effect of the mapping strategy itself [16]. Therefore, a direct comparison between PCA and AE within the same BoMI framework and under the same experimental conditions is needed. In this study, we compare a linear PCA-based mapping and a nonlinear AE-based mapping for teleoperation of a simulated mobile robot base. Hand motion was captured using a standard webcam and represented as a 42-dimensional vector of hand landmark coordinates, which was mapped into a two-dimensional control space. The resulting cursor position was then converted into linear and angular velocity commands for the robot base. We hypothesized that, under strong dimensionality reduction, the nonlinear AE mapping would provide more efficient and stable teleoperation than the linear PCA mapping.

II. METHODS

A. Experimental Apparatus

The overall architecture of the proposed BoMI is shown in Fig. 1. The interface was implemented using custom Python scripts and consisted of three main modules:

1) Data Acquisition and Processing: The acquisition and processing module of the BoMI, illustrated in Fig. 1(A), was based on previous work from our group [17] and used a standard PC webcam together with MediaPipe [18]. The webcam recorded the motion of one hand in real time. The system acquired video frames using the OpenCV library and processed them through MediaPipe to detect 21 hand landmarks. It then concatenated the 2D image coordinates (x, y) of these landmarks to form a 42D hand-motion vector, which it updated at the video sampling rate of 50 Hz.

2) Data Mapping: The data mapping module transformed the hand-motion coordinates into cursor movements through a Finger-to-Cursor Mapping. Specifically, the 42D hand landmark vector q was mapped into a 2D control vector p, representing the cursor coordinates on the screen. Two mapping strategies were implemented and compared: a linear mapping based on PCA and a nonlinear mapping based on an AE. The same calibration dataset, consisting of 90 s of free hand movements, was used to compute both mappings, allowing a direct comparison between the two approaches under equivalent input conditions. For the linear approach, PCA was applied as in [1]. The mapping was defined as:

$$p_s = S_s * (H * q) + p_0 \tag{1}$$

where $q \in \mathbb{R}^{42}$ is the hand-landmark vector, $H \in \mathbb{R}^{2\times42}$ is the PCA-based transformation matrix, $S_s \in \mathbb{R}^{2\times2}$ is a subject-specific scaling matrix, $p_0 \in \mathbb{R}^2$ is a subject-specific offset vector, and $p_s \in \mathbb{R}^2$ is the resulting 2D cursor position for participant s. The two principal components (PCs) associated with the largest eigenvalues were used to project the 42D hand-motion vector into the 2D control space. Once computed, the PCA mapping was kept fixed for all participants assigned to the PCA group. For the nonlinear approach, an AE was used [7]. The AE consisted of encoder and decoder subnetworks. The encoder compressed the 42D input vector into a bottleneck layer, while the decoder reconstructed the original input from the latent representation. The encoder subnetwork was used as the forward mapping from the hand-motion space to the 2D control space. The bottleneck layer contained two coding units (CUs), which generated the 2D cursor position. In the PCA- and AE-based BoMI approaches, the first PC or CU controlled the cursor displacement along the horizontal axis, while the second PC or CU controlled the cursor displacement along the vertical axis.

3) Control Interface: The Mobile Base Control Interface allowed participants to teleoperate the robot base through a Nine-Region Graphical User Interface, shown in Fig. 1(D). This interface acted as a virtual joystick: the cursor workspace was divided into nine regions, each associated with a specific movement command for the virtual mobile robot base. Cursor coordinates were converted into linear and angular velocity commands based on the region in which the cursor was located. Vertical cursor displacement controlled forward and backward motion, whereas horizontal displacement controlled left and right turning. The resulting velocity commands were streamed through ROS to the Gazebo simulation environment at the same update rate as the hand landmark extraction, namely 50 Hz.

B. Validation of the AE

The validation procedure was designed to identify the architecture and hyperparameters of the AE that could provide an effective forward map for robot-base control. Specifically, the selected AE was required to satisfy two criteria: (i) low information loss during dimensionality reduction, and (ii) balanced distribution of variance across its two CUs. The validation was performed using a dataset collected from one subject, who did not participate in the experimental study. The subject was a 30-year-old male who was instructed to freely move the left hand for 90 s while keeping the forearm supported and secured in the hand holder with a strap, as shown in Fig. 1(A). The resulting dataset had 4500 samples, corresponding to 90 s of acquisition at a sampling rate of 50 Hz, and was then divided into 3600 training samples and 900 test samples, corresponding to an 80/20 split. The AE architecture and hyperparameters were tuned following the procedure described in [7]. Two candidate AE architectures were evaluated. First, an *under-parameterized AE* was trained. This network had fewer trainable parameters than the number of training samples. It consisted of four hidden layers, two in the encoder and two in the decoder, with eight neurons per layer, resulting in 908 trainable parameters. Second, an *over-parameterized AE* was trained. It had the same general structure, with two hidden layers in the encoder and two in the decoder, but each hidden layer contained 32 neurons, resulting in 5036 trainable parameters. To quantify the amount of information preserved by each AE, we computed the Variance

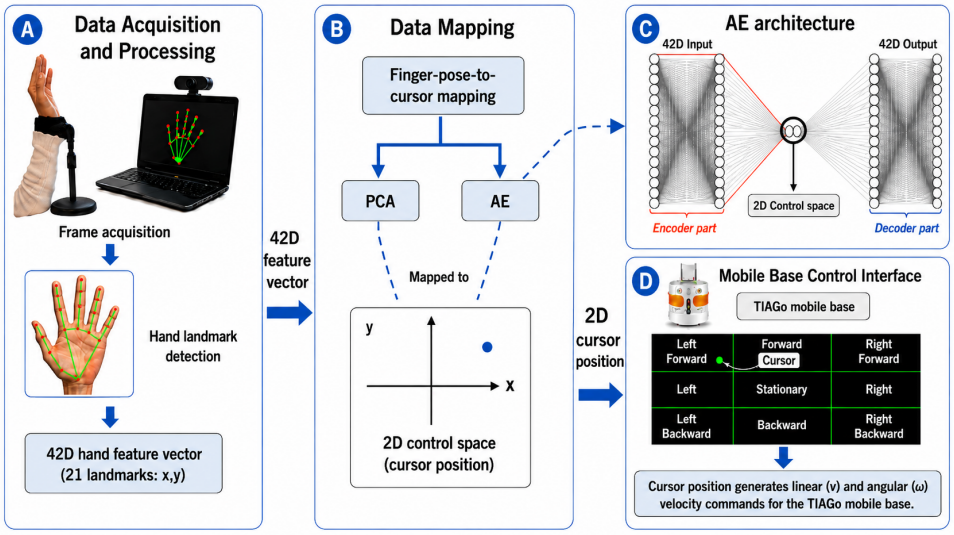


Fig. 1. Body–Machine Interface architecture, structured into the following the modules: data acquisition and processing (Panel A), data mapping (Panel B), AE architecture (Panel C), and mobile-base control interface (Panel D).

Accounted For (VAF), defined as:

$$\mathrm{VAF} = \left(1 - \frac{\mathrm{var}(q - \hat{q})}{\mathrm{var}(q)}\right) \times 100 \tag{2}$$

where q is the original dataset and $\hat{q}$ is the dataset reconstructed by the AE. In addition, we evaluated how the variance was distributed across the AE latent dimensions. For each CU, the latent variance was computed as:

$$v_{\mathrm{lat},i} = \frac{v_i}{\sum_{j=1}^{2} v_j} \times 100 \tag{3}$$

where v_i is the variance of the i-th CU. The training objective was to select an AE that achieved high reconstruction accuracy while also producing a balanced representation across the two latent dimensions, corresponding to approximately 50% of the variance in each CU. For comparison, PCA was also applied to the same dataset. Based on this validation, the over-parameterized AE was selected as the final architecture with its encoder subnetwork used as the BoMI forward map for online control of the robot base. The encoder transformed the 42D hand-motion vector q into the 2D control vector θ according to:

$$\begin{aligned} layer_1 &= \tanh(w_1 * q + b_1) \\ layer_2 &= \tanh(w_2 * layer_1 + b_2) \\ \theta &= w_3 * layer_2 + b_3 \end{aligned} \tag{4}$$

where $\tanh$ is the nonlinear hyperbolic tangent activation function used in each hidden layer, and w_j and b_j, for $j = 1, 2, 3$, are the weights and biases of the encoder subnetwork, respectively.

C. Participants

Fourteen unimpaired participants were recruited to evaluate online control of the virtual robot base. All participants were right-handed, except for one participant who was left-handed. Participants were randomly assigned to one of the two groups: the AE group, composed of seven participants (3 females, age 26.86 ± 3.63 years), and the PCA group, composed of seven participants (2 females, age 25.29 ± 2.43 years). Participants reported no known history of postural, musculoskeletal, or neurological disorders and showed normal joint range of motion and muscle strength. Within each group, all participants used the same mapping during the experiment. Specifically, the PCA group used the PCA-based map computed once from the calibration dataset, whereas the AE group used the encoder subnetwork of the selected AE, which was trained and validated using the same calibration dataset. The experimental protocol was approved by the DIBRIS Institutional Review Board, University of Genoa (Protocol No. 009/2020; May 18, 2020).

D. Experimental Protocol

During the experiment, participants sat in front of two computer displays. The first was a laptop with an integrated webcam, which was used for video acquisition and visual feedback through the control interface. The second display presented the simulated mobile-base environment. The participant's wrist was positioned inside a holder with a strap, as shown in Fig. 1(A), to maintain a stable posture during the experiment. A reference mask, based on the hand configuration used during the calibration procedure, was displayed to guide hand placement. After the participant's wrist was secured, the laptop and webcam were adjusted to align the hand with the reference mask. This procedure was used to ensure consistent hand positioning and scaling relative to the camera across participants. The experimental protocol consisted of a familiarization phase, followed by an initial test block, two training blocks, a mid test block, two additional training blocks, and a final test block (see Fig. 2(A)). During the familiarization phase, participants practiced controlling the

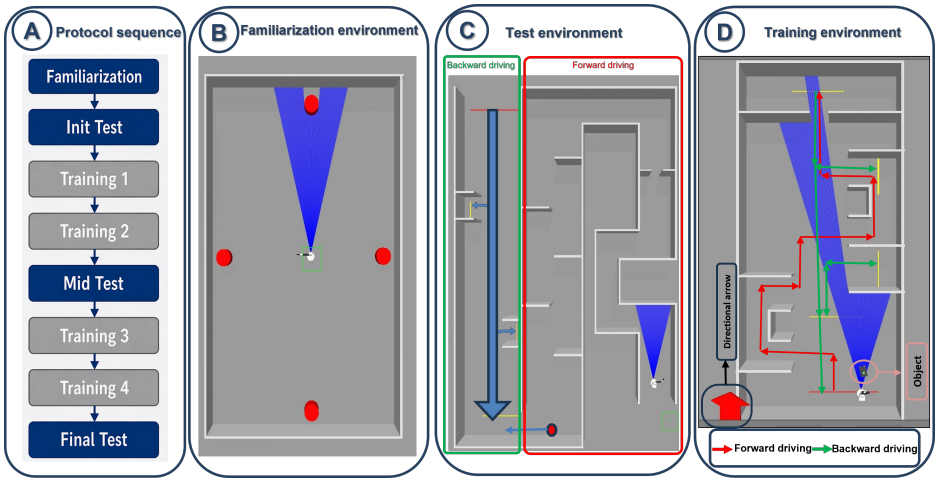


Fig. 2. Experimental protocol and simulated navigation environments. The protocol sequence is shown in Panel A. Panel B shows the familiarization environment, Panel C shows the test environment with forward- and backward-driving sections, and Panel D shows the training environment with forward and backward driving paths.

robot in a bounded open environment enclosed by four walls, as shown in Fig. 2(B). The map contained four red cylindrical targets positioned in different directions around the robot. Participants were given 3 minutes to freely navigate the room, teleoperating the robot in multiple directions and attempting to reach any of the red cylinders. This phase was intended to familiarize the participants with the control interface and the robot's motion responses before the main experimental tasks. In the test blocks, participants controlled the robot in a map that included two navigation modes. The first mode consisted of forward driving, corresponding to the region highlighted by the red frame in Fig. 2(C). Participants started from the initial position and drove the robot through the complete forward-driving route, which included straight segments, curves, and a slalom section, before transitioning to the second mode. The second mode consisted of backward driving, corresponding to the region highlighted by the green frame in Fig. 2(C). In this section, participants drove the robot in reverse through straight paths and turns. The test task ended when the robot reached the red line. The same map was used consistently across all test blocks. In the training blocks, participants controlled the robot in a structured environment, different from the one used in the test block, composed of straight segments, curved sections, and turns. The task required two navigation modes: forward driving and backward driving. As shown in Fig. 2(D), forward-driving segments are indicated by red arrows, whereas backward-driving segments are indicated by green arrows. The robot started from the red line and was required to return to the same line at the end of each training task. A square-shaped virtual cart was displayed as a reference trajectory for participants to follow, guiding them along the prescribed path. In addition, a large directional arrow was shown on the screen to indicate whether forward or backward driving was required. The same map was used consistently across all training blocks. The full protocol was completed in a single session. No maximum time limit was imposed for completing the overall experiment.

E. Outcome Metrics

Four outcome metrics were computed to evaluate participants' performance during the online BoMI test blocks:

- **Completion time:** the time required to successfully complete each test block. A decrease in completion time across test blocks was interpreted as an improvement in task performance.
- **Robot path length:** the total distance traveled by the robot from the starting position to the endpoint during each test block. Shorter path lengths indicate more efficient navigation.
- **Number of control-interface region transitions:** the number of transitions made by the cursor between regions of the control interface. A lower number of transitions indicates more consistent and stable control behavior.
- **Wall collisions:** the number of times the robot collided with the walls during each test block. A reduction in collisions indicates safer and more accurate robot navigation.

These metrics were calculated for the initial, mid, and final test blocks.

F. Statistical Analysis

For each outcome metric, we used a mixed repeated-measures ANOVA with *block* as the within-subject factor (initial, mid, and final test) and *method* as the between-subject factor (PCA and AE). Mauchly's test was used to assess sphericity, and Greenhouse–Geisser correction was applied when required. Post hoc pairwise comparisons were performed when appropriate. The significance level was set at $p < 0.05$.

III. RESULTS

A. Validation of the AE

Using a two-dimensional latent space, PCA and the over-parameterized AE preserved a comparable amount of information from the calibration dataset, with both methods achieving

a VAF of 78% (Fig. 3(A)). In contrast, the under-parameterized AE retained a lower percentage of the calibration variance, with a VAF of 57% (Fig. 3(A)). For the AE models, this comparison was performed using an initialization that produced a similarly balanced CU distribution in both architectures. Under this comparable latent-space condition, the number of trainable parameters affected reconstruction performance, as shown by the difference in VAF between the under-parameterized and over-parameterized architectures. However, the distribution of variance across the latent dimensions was not substantially affected by the number of parameters (Fig. 3(B)). Both AE architectures showed an approximately uniform distribution of variance across the two code units, with each control unit accounting for nearly half of the latent variance. In contrast, the PCA-based mapping produced an imbalanced variance distribution with the first principal component accounting for 70% of the control-space variance (Fig. 3(B)). Based on these results, the over-parameterized AE was selected for the online BoMI test.

B. Online Control Performance

With practice, participants improved their performance across the three test blocks. Specifically, completion time significantly decreased from the initial to the final test block ($F(2,24) = 33.16$, $p < 0.001$, $\eta_p^2 = 0.734$, Fig. 4(A)). The AE group completed the task significantly faster than the PCA group ($F(1,12) = 28.7$, $p < 0.001$, $\eta_p^2 = 0.705$), and a significant block-by-method interaction was found ($F(2,24) = 9.73$, $p < 0.001$, $\eta_p^2 = 0.448$). The largest difference between groups was observed during the initial test block, where participants using the AE mapping completed the task significantly faster than those using PCA (post hoc comparison, $p < 0.001$, Cohen's $d = 2.72$). This difference remained evident in the final test block, although it was smaller than in the initial test block (post hoc comparison, $p = 0.007$, Cohen's $d = 1.75$). Robot path length decreased significantly across test blocks ($F(2,24) = 7.13$, $p = 0.004$, $\eta_p^2 = 0.373$). Post hoc comparisons showed reductions from the initial to the mid test (Cohen's $d = 0.81$) and from the initial to the final test (Cohen's $d = 0.68$). However, there was no significant effect of mapping method or block-by-method interaction. The number of transitions between regions of the control interface decreased significantly across test blocks ($F(2,24) = 9.49$, $p < 0.001$, $\eta_p^2 = 0.442$, Fig. 4(C)). The AE group also performed significantly fewer transitions than the PCA group ($F(1,12) = 24.1$, $p < 0.001$, $\eta_p^2 = 0.668$), with a significant block-by-method interaction ($F(2,24) = 4.98$, $p = 0.015$, $\eta_p^2 = 0.293$). Post hoc comparisons showed that this difference was largest during the initial test block ($p < 0.001$, Cohen's $d = 2.72$) and remained present in the final test block ($p = 0.015$, Cohen's $d = 1.52$). These results suggest that the AE mapping supported more stable and consistent cursor control than PCA. Finally, the number of wall collisions decreased significantly across test blocks ($F(2,24) = 3.90$, $p = 0.034$, $\eta_p^2 = 0.245$, Fig. 4(D)). Post hoc analysis showed a significant reduction from the initial to the final test block ($p = 0.040$, Cohen's $d = 0.71$). The AE group also showed significantly fewer collisions than the PCA group ($F(1,12) = 5.74$, $p = 0.034$, $\eta_p^2 = 0.323$). Overall, these results indicate that both groups improved with practice, but the AE-based BoMI provided faster task completion, fewer control-interface transitions, and safer navigation than the PCA-based BoMI.

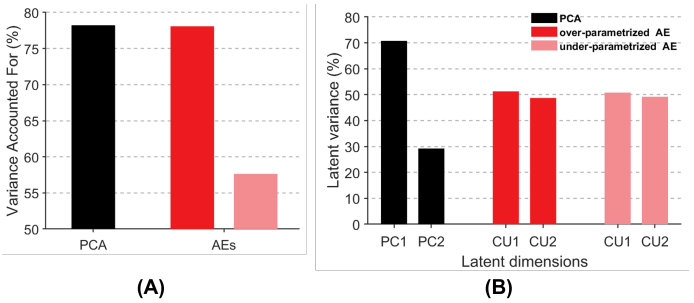


Fig. 3. Validation of the dimensionality-reduction mappings. Panel A shows the Variance Accounted For (VAF) by PCA and the AE architectures. Panel B shows the distribution of variance across the two latent dimensions for PCA, the over-parameterized AE, and the under-parameterized AE.

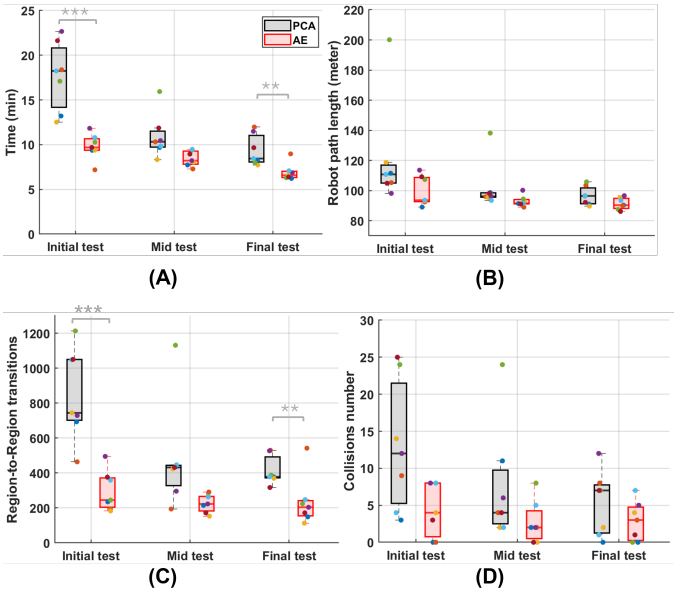


Fig. 4. Online control performance during the test blocks. Performance is shown for the PCA and AE groups across the initial, mid, and final tests: completion time (A), robot path length (B), number of control-interface region transitions (C), and number of wall collisions (D). $^{**}p < 0.01$, $^{***}p < 0.001$.

IV. DISCUSSION

The aim of this study was to investigate whether nonlinear mappings based on AEs provide a practical advantage over conventional linear approaches, such as PCA, during navigation-like teleoperation tasks under strong dimensionality reduction. Previous studies have shown that body-motion signals can be used to control cursors, wheelchairs, and robotic systems [1], [6], [12]–[15], but the specific effect of the mapping on the quality of the control remains unclear,

especially for mobile-base navigation. The validation results showed that the AE and PCA retained a comparable amount of the calibration variance. The lower VAF observed for the under-parameterized AE should be interpreted within the validation procedure, where the selected initialization was chosen to provide comparable CU balance across AE architectures. However, as expected, PCA produced an imbalanced variance distribution across the two PCs, whereas the AE distributed variance more uniformly across the two CUs. Since the proposed interface required both control dimensions to generate the cursor position and transform this position into linear and angular velocity commands, this more balanced latent space may have supported more stable cursor control. The online results showed that both groups improved with practice, confirming that participants could learn to control the robot with either mapping. However, the AE group completed the task faster, made fewer transitions between control-interface regions, and had fewer wall collisions than the PCA group. These findings suggest that the AE mapping improved not only task speed, but also command stability and navigation safety. This extends previous work on video-based BoMIs [17] and nonlinear BoMIs for robotic control [7] by showing that nonlinear mappings can be beneficial for mobile-base teleoperation. In our case, this advantage cannot be explained simply by a higher amount of preserved information, because PCA and the selected AE retained a comparable percentage of the calibration dataset variance. Rather, the main difference appeared to be the structure of the resulting control space: PCA concentrated most of the variance in the first principal component, 70%, whereas the AE produced a more balanced distribution across the two code units, with each code unit accounting for 50% of the latent variance. Since both control dimensions were required to generate the cursor position and subsequently the linear and angular velocity commands, this more balanced latent space may have reduced the dominance of one control axis and supported more stable command generation. Therefore, AE-based BoMIs represent a promising approach for developing intuitive and effective interfaces for continuous robot teleoperation. Although the increased sample size strengthens the findings, validation in larger and more diverse populations is still required. This study represents a first simulation step toward real-world deployment of the proposed BoMI framework. Future work will transfer the control system to the Reachy robot and evaluate it in real-world mobile-base navigation tasks.

V. CONCLUSION

This study compared PCA- and AE-based mappings for webcam-based BoMI control of a simulated mobile robot base. Both mappings supported learning, but the AE mapping produced faster, more stable, and safer navigation, suggesting that nonlinear mappings are advantageous under strong dimensionality reduction.

REFERENCES

[1] M. Casadio, A. Pressman, A. Fishbach, Z. Danziger, S. Acosta, D. Chen, H.-Y. Tseng, and F. A. Mussa-Ivaldi, "Functional reorganization of upper-body movement after spinal cord injury," *Experimental brain research*, vol. 207, no. 3, pp. 233–247, 2010.

[2] C. Antfolk, C. Cipriani, M. Controzzi, M. C. Carrozza, G. Lundborg, B. Rosén, F. Sebelius *et al.*, "Using emg for real-time prediction of joint angles to control a prosthetic hand equipped with a sensory feedback system," *Journal of Medical and Biological Engineering*, vol. 30, no. 6, pp. 399–406, 2010.

[3] D. W. Franklin and D. M. Wolpert, "Computational mechanisms of sensorimotor control," *Neuron*, vol. 72, no. 3, pp. 425–442, 2011.

[4] A. Ajoudani, N. Tsagarakis, and A. Bicchi, "Tele-impedance: Teleoperation with impedance regulation using a body–machine interface," *The International Journal of Robotics Research*, vol. 31, no. 13, pp. 1642–1656, 2012.

[5] D. J. Rea and S. H. Seo, "Still not solved: A call for renewed focus on user-centered teleoperation interfaces," *Frontiers in Robotics and AI*, vol. 9, p. 704225, 2022.

[6] F. Abdollahi, A. Farshchiansadegh, C. Pierella, I. Seáñez-González, E. Thorp, M.-H. Lee, R. Ranganathan, J. Pedersen, D. Chen, E. Roth *et al.*, "Body-machine interface enables people with cervical spinal cord injury to control devices with available body movements: Proof of concept," *Neurorehabilitation and neural repair*, vol. 31, no. 5, pp. 487–493, 2017.

[7] F. Rizzoglio, M. Giordano, F. A. Mussa-Ivaldi, and M. Casadio, "A non-linear body machine interface for controlling assistive robotic arms," *IEEE Transactions on Biomedical Engineering*, vol. 70, no. 7, pp. 2149–2159, 2023.

[8] F. Rizzoglio, C. Pierella, D. De Santis, F. Mussa-Ivaldi, and M. Casadio, "A hybrid body-machine interface integrating signals from muscles and motions," *Journal of Neural Engineering*, vol. 17, no. 4, p. 046004, 2020.

[9] A. Freccero, M. Feder, G. Grioli, M. G. Catalano, A. Massone, A. Bicchi, and M. Casadio, "A body–machine interface for assistive robot control in spinal cord injury: system description and preliminary tests," *Applied Sciences*, vol. 15, no. 4, p. 1792, 2025.

[10] M. Macchini, J. Frogg, F. Schiano, and D. Floreano, "Does spontaneous motion lead to intuitive body-machine interfaces? a fitness study of different body segments for wearable telerobotics," in *2022 31st IEEE International Conference on Robot and Human Interactive Communication (RO-MAN)*. IEEE, 2022, pp. 115–121.

[11] C. Pierella, C. D'Antuono, G. Marchesi, C. E. Menotti, and M. Casadio, "A computer interface controlled by upper limb muscles: effects of a two weeks training on younger and older adults," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 31, pp. 3744–3751, 2023.

[12] M. Casadio, R. Ranganathan, and F. A. Mussa-Ivaldi, "The body-machine interface: a new perspective on an old theme," *Journal of Motor behavior*, vol. 44, no. 6, pp. 419–433, 2012.

[13] X. Wang, M. Casadio, K. A. Weber II, F. A. Mussa-Ivaldi, and T. B. Parrish, "White matter microstructure changes induced by motor skill learning utilizing a body machine interface," *Neuroimage*, vol. 88, pp. 32–40, 2014.

[14] C. Pierella, F. Abdollahi, E. Thorp, A. Farshchiansadegh, J. Pedersen, I. Seáñez-González, F. A. Mussa-Ivaldi, and M. Casadio, "Learning new movements after paralysis: Results from a home-based study," *Scientific reports*, vol. 7, no. 1, p. 4779, 2017.

[15] E. B. Thorp, F. Abdollahi, D. Chen, A. Farshchiansadegh, M.-H. Lee, J. P. Pedersen, C. Pierella, E. J. Roth, I. S. Gonzáles, and F. A. Mussa-Ivaldi, "Upper body-based power wheelchair control interface for individuals with tetraplegia," *IEEE transactions on neural systems and rehabilitation engineering*, vol. 24, no. 2, pp. 249–260, 2015.

[16] S. N. Young and J. M. Peschel, "Review of human–machine interfaces for small unmanned systems with robotic manipulators," *IEEE Transactions on Human-Machine Systems*, vol. 50, no. 2, pp. 131–143, 2020.

[17] M. Moro, F. Rizzoglio, F. Odone, and M. Casadio, "A video-based markerless body machine interface: A pilot study," in *International Conference on Pattern Recognition*. Springer, 2021, pp. 233–240.

[18] C. Lugaresi, J. Tang, H. Nash, C. McClanahan, E. Uboweja, M. Hays, F. Zhang, C.-L. Chang, M. Yong, J. Lee *et al.*, "Mediapipe: A framework for perceiving and processing reality," in *Third workshop on computer vision for AR/VR at IEEE computer vision and pattern recognition (CVPR)*, vol. 2019. Long Beach, CA, 2019, p. 2.